

Adithya Udupa

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Experience

Software Engineer, Intern - Agilysys

Jun 2026 – Present

- **Engineered AI Layout Pipeline:** Built an end-to-end AI digital menu board layout engine, similar to Canva, using React (Vite) and Node.js (Hono), reducing manual board creation time to under 90 seconds.
- **Integrated Generative AI:** Leveraged Flux Pro and GPT-4 Vision to convert prompts into validated JSON schemas, generating layouts across 1080p and 4K displays.
- **Developed POS Data Binding:** Built RESTful APIs to integrate Point-of-Sale menu data directly into the board generator, successfully importing live menus without relying on manual drag-and-drop intervention.
- **Delivered Tested MVP:** In 3 months, shipped a full-stack prototype supported by around 200 automated unit tests and audit logging for all AI generations.

Systems and Level Designer - Mars Games

Oct 2024 – Apr 2026

- **Delivered 8 complete Points of Interest (POIs) across multiple biomes**, including the main new-player landing area, the Train Station, by utilizing Unreal Engine 5 to establish core gameplay loops.
- **Enhanced environmental interactivity and player challenge** by designing, scripting, and implementing multiple biome-specific hazards (e.g., sinking logs, broken bridges, trap doors).
- **Expanded combat dynamics** by researching, prototyping, and documenting distinct enemy AI behaviors and custom combat systems, both melee and ranged.
- **Achieved finalized visual and functional aesthetics** by maintaining comprehensive architectural and asset documentation in Jira and Confluence while cross-collaborating with producers and art teams.
- **Improved overall combat balance** by creating structured tracking documentation to monitor and tune enemy AI parameters alongside player combat metrics.

Projects

Unannounced AAA Multiplayer Game - Level designer, systems designer (intern)

- **Tools Used:** Unreal Engine 5, Blender, Blueprints, Confluence, Jira, Photoshop.
- **Led the spatial design and environmental layout** for 8 distinct Points of Interest (POIs), including the critical main spawn point, by utilizing Unreal Engine 5.
- **Worked on core gameplay mechanics** by implementing blueprint logic in cross-functional collaboration with the engineering and systems design teams.

Ready Or Not Mod - Mapper, level designer, environment artist, lighting artist

- **Tools Used:** Unreal Engine 5, Blender, Blueprints, Github, Photoshop.
- **Established highly replayable tactical encounters** for the published custom map *Cost Of Living* by designing complex architectural layouts, intuitive golden paths, and comprehensive player navigation systems.
- **Ensured seamless and responsive NPC behavior** by implementing precise AI navigation bounds, including specific room volumes, dynamic doors, SWAT stack-up points, and logical clearing routes.
- **Published on Nexus Mods** with 40'000 views, 17'000 downloads, and YouTube features for custom maps.

Daredevil, Solo Project - Systems designer, level designer, UI designer, UI artist, and environment artist

- **Tools Used:** Unreal Engine 5, Blender, GitHub, GIMP, Photoshop.
- **Developed core character mechanics**, like Radar sense, parkour traversal, and hand-to-hand combat, by programming custom logic and blueprints in Unreal Engine 5.
- **Implementing a scalable and modular combat architecture** by using the Gameplay Ability System (GAS) to systematically drive complex gameplay attributes, status effects, and ability events.
- **Delivered a cohesive and responsive player experience** by designing, planning, and integrating custom heads-up displays (HUDs) and front-end user interfaces utilizing GIMP and Photoshop. and events.

Elixir Blitz: Saya's Trials, Group Project - Level designer, UI artist, UI designer, and environment artist

- **Tools Used:** Unreal Engine 5, Blender, Diversion, GIMP, Photoshop.
- **Engineered the complete gameplay flow for the game's second level** by conceptualizing, planning, and designing the full environmental layout within Unreal Engine 5.
- **Streamlined the player experience** by designing and integrating a suite of functional user interfaces, like the main menu, pause screen, settings, and interactive inventory, using GIMP, Photoshop, and UE5 blueprints.

Skills

Programming: React | Node.js | UE5 | Blueprints | Unity | SQL | C++ | C# | Python | Kotlin

Production and QA: Jira | Confluence | Agile Methodology | MS Office & Google Suites | GitHub

Design & Art: Blender | Photoshop | GIMP | Maya | ZBrush | Adobe CC | Aseprite

Education

Full Sail University | *Master of Science in Game Design.*

Oct 2025

- Coursework: Advanced Production Techniques, Game Design, Asset Management, QA, UX Research.

RV Institute of Technology | *Bachelor of Engineering in Computer Science*

Aug 2023

- Coursework: C++, C, SDLC, OOPs, Big Data Analysis, Data Mining, Python, AI/ML, UI/UX.

Additional Information

Languages: English (IELTS C2), Kannada, Japanese, Hindi.